

SENSING SOUNDS

COCHLEAR IMPLANTS

Hearing aids can help people who cannot hear well by amplifying the sounds in their ear. However, some people have such profound hearing loss that this doesn't work. Cochlear implants are an invention designed to bypass the inner ear by converting sound into electrical signals, which are then transmitted directly into the brain. One part is surgically implanted and the other sits outside the body on the side of the head.



SEEING BY SENSORS

Bionic eyes work in a similar way to cochlear implants and are designed to help people with sight loss, although they are not yet widely available. They use a camera mounted on glasses to pick up visual information and send it directly to the brain.

The brain receives the signals and processes them.

THE EAR

Hearing is simply our ear and brain interpreting vibrations in the air. When sound waves reach our ear, they make the eardrum vibrate. These vibrations travel through tiny bones to the cochlea, which converts them into signals for the brain.

